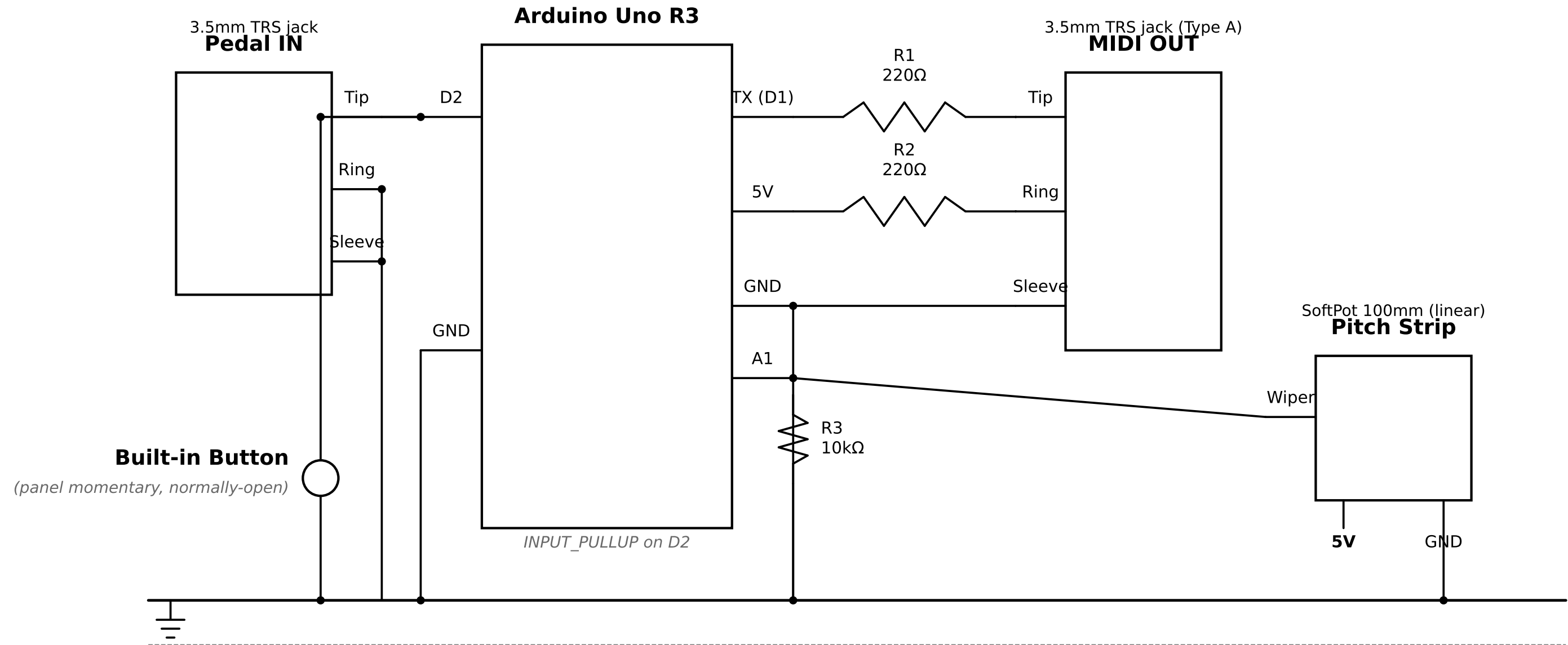


# FM-1 Sustain Footswitch — Schematic

Arduino Uno R3 + TRS Pedal In + Built-in Button + TRS MIDI Out (Type A) + Softpot Pitch Strip



## Notes:

- Built-in Button is wired in parallel with the pedal jack (same D2/GND nodes) -- either one alone can trigger sustain, so an external pedal is optional
- Pedal ring + sleeve are tied together so a plain mono (TS) pedal plug still grounds correctly in the TRS jack
- D2 uses INPUT\_PULLUP -- the button or pedal only needs to short D2 to GND when pressed (normally-open momentary)
- Type A TRS MIDI: tip = signal, ring = +5V current source, sleeve = ground
- R1 limits current through the FM-1's opto-isolated MIDI input; R2 sources the +5V loop current -- both 220Ω, matching the standard DIN MIDI-out circuit
- TX (D1) is shared with the USB-serial programmer -- disconnect the MIDI OUT jack (or at least the R1 lead) before uploading a new sketch
- If the FM-1 doesn't respond, it may expect Type B instead: swap Tip and Ring at the MIDI OUT jack (tip = GND, ring = signal, sleeve = GND)
- Pitch Strip: End 1 -> 5V, End 2 -> GND, Wiper -> A1 through R3 (10kΩ pull-down) -- base note sounds continuously from power-on, sliding the strip bends pitch, releasing settles back to center (no bend); no button needed
- Over the FM-1's physical MIDI IN (not USB), continuous Pitch Bend applied to an already-sounding note works and sounds smooth -- confirmed 2026-08-06, no retrigger-glide workaround needed on this path (that was only required for the FM-1's USB MIDI input)
- OLED display, mode switch, and audio-input pitch detector removed -- the OLED moved permanently to a separate project, fm1-midi-voice-tuner